What is Situated AI?

Von Neumann Architecture Unconsciously Shapes Expectations of Intelligence

Situated AI is at least 30 years old. Yet despite its dominance of practical robotics since the 1990s, its ideas remain controversial and its successes obscured by the more easily laudable achievements of traditional AI. Why?

Von Neumann Architecture

The assumptions of Von Neumann Architecture (VNA) have unconsciously shaped expectations of intelligent behavior for decades: it's generally assumed that a *central control unit* is necessary and inevitable. These assumptions prevail:

1. Without global coordination, chaos must result.
2. Without abstract representation, navigation and search are impossible.
3. Without symbolic reasoning, higher intelligence is impossible.

Situated emergent AI defies these expectations: since the late 1980s, the AI researchers centered around the work of Rodney Brooks and his MIT laboratory have proven the viability of the *bottomup approach* to intelligence.

> Almost all Artificial Intelligence research, and indeed almost all modern computation, runs on essentially Von Neumann architectures, with a large, inactive memory which can respond at very high speed over an extremely narrow channel, to a very high speed central processing unit which contains very little state.
>
> *Intelligence Without Reason*, §3

Let's recall what VNA is. What are its components?

1. Control Unit (CU): translates incoming external instructions into steps allocated to the CPU, memory registers, and output. Important subunits are the instruction register and instruction pointer.

2. Central Processing Unit (CPU): Important subunits are the Arithmetic Logic Unit (ALU) and processor register.

3. Global shared memory: stores data and instructions. Implemented as random access memory (RAM) in most modern computers.

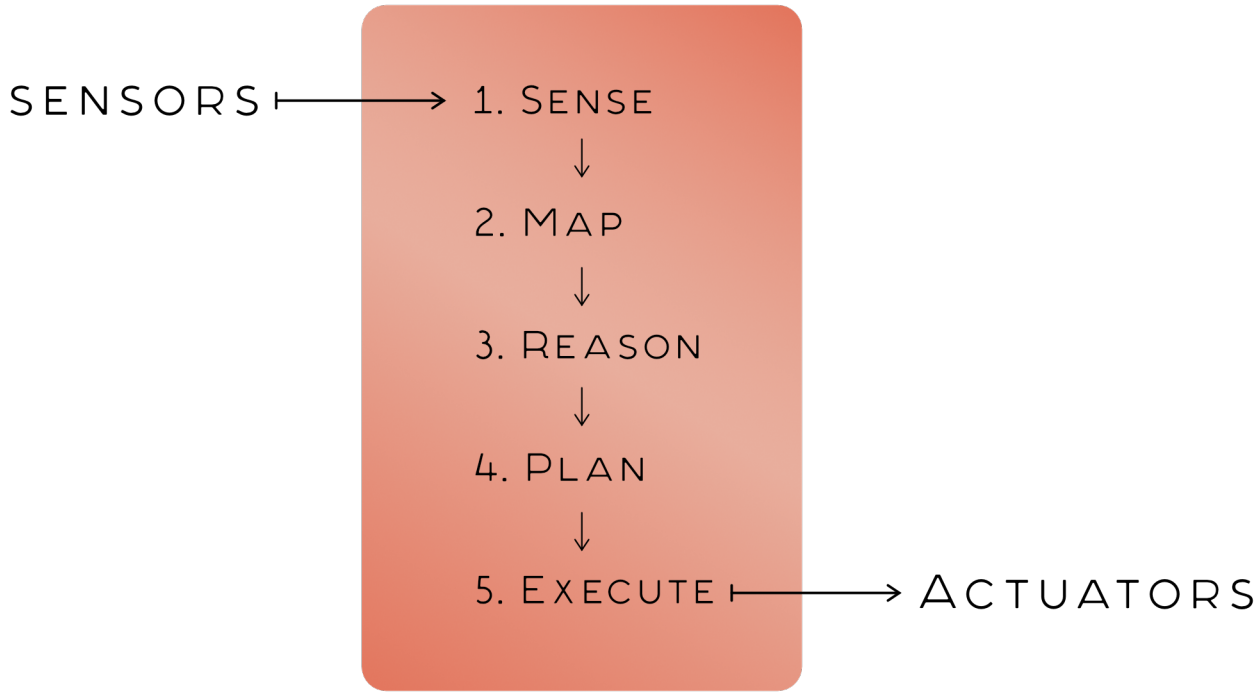
4. External mass storage, over a high latency bus and thus rarely accessed during processing.

5. High-speed, narrow input and output channels: an internet connection as input and an LED monitor as output constitute the most familiar examples.

A VNA computer is essentially a Boolean calculator strapped to a resource allocator optimized around the limitations of the design. Synchronous processors spend an enormous amount of time waiting. Despite recent emphasis on concurrency and the exploitation of parallel graphics processors, parallelism implemented in VNA still remains an afterthought.

Situated vs. Traditional AI

What is "Situated AI"? It is intelligent behavior evoked from a stack of robustly competent, overlapping, asynchronous behaviors. In situated AI, there is no central processor, no shared global memory, no central representation of the world, as little symbolic reasoning as possible, and as much independence and modularity between behaviors as possible.



The signature of situated AI, is the *vertical stack* of independent modules, each with their own access to input parameters and output behaviors, such that every lower level is more *independent* and *agnostic* than the one above it. The upper layers may monitor the data of lower layers and selectively interfere in their output streams, without the knowledge of the lower layer, as long as the primary goal of the lower layer is not violated: such as self-preservation, human safety, remaining upright, and so on.

The argument is that this design more closely resembles biological systems, which are massively parallel and modular: for example, at the cellular level there is no central "control" cell. Transcription and metabolism proceeds in each cell, both independent of its neighbors and profoundly influenced by the extracellular environment, which is full of overlapping and orthogonal channels of

communication. Even within one well-defined system with relatively clear goals, such as the immune system, there are many overlapping subsystems and channels of transcription and feedback: the several interferon channels, the histaminic, etc. The result is functional redundancy and robust error recovery: should a pathogen defeat one of those subsystems, another will take its place and eventually eliminate the pathogen even at reduced capacity.

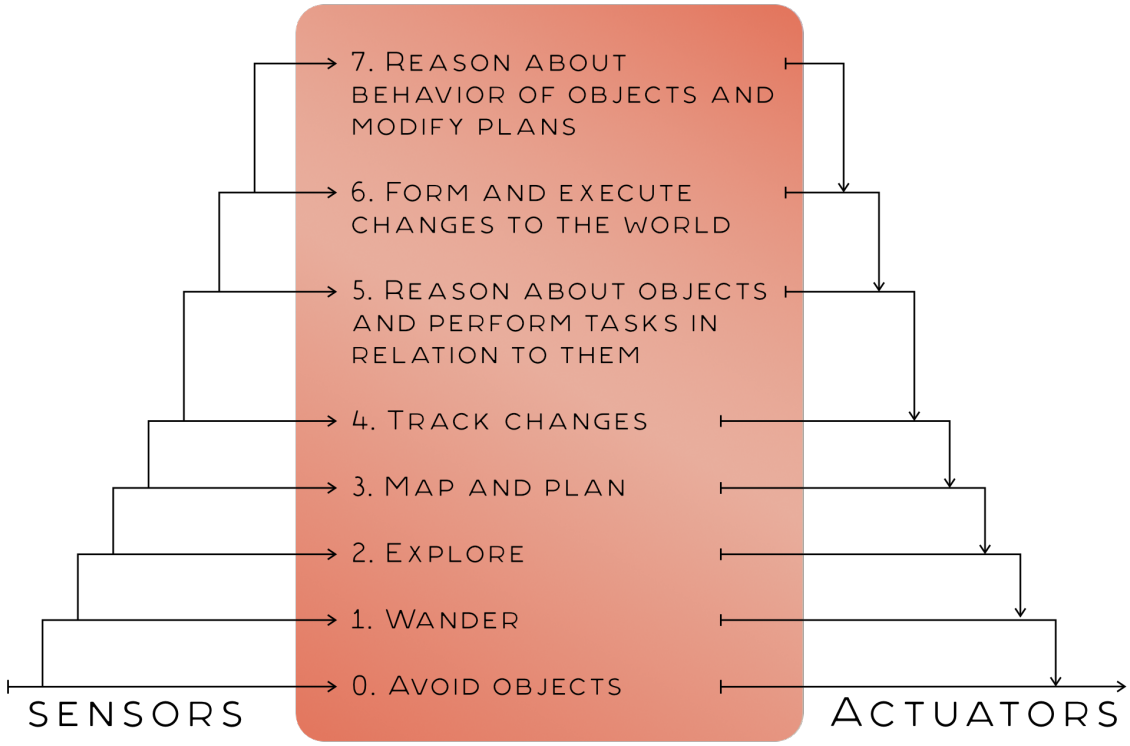
Traditional AI has been modeled after conventional ideas concerning *conscious* behavior and its relationship to intelligence. Therefore AI research has been frequently limited by folk psychology and *naïve agency modeling*. As though to achieve intelligent behavior, an organism must follow these steps sequentially:

1. Sense
2. Map
3. Reason
4. Plan
5. Execute

But as ethology and neuroscience have shown for many years, this is nothing like the behavior of a living creature. We have ample evidence that a biological system is functionally redundant and modular: demonstrated by the many case studies collated by neuroscientists like Oliver Sacks, alongside the many vivid examples of functional survivors of traumatic injury, such as the famous case of Phineas Gage. Independence between systems is the norm, not the exception.

Concurrent Agnostic Competence

Situated AI attempts to generate this kind of robust parallelism by building "levels of competence", with every successive level more specific and constrained. Yet the lower layers should operate regardless of the interference or activity of the upper layers, asynchronously and agnostically.



What does it mean, when Brooks says, “Time is its own representation”, or “The world is its own best model”?

The idea is that rather than attempting to represent quantities and measurements in a virtual symbolic language, and then reason about them using cleverly designed algorithms, those quantities are allowed to percolate through the system and determine behavior directly. For example, a time value can be “represented” simply by a delay between two signals, and “reasoned” about by effecting the output of a behavior sensitive to signal dispersion. Those behaviors don’t need to “know” what time is, to act in a *timely* manner: in fact such representation almost invariably tends to produce slow responses, generating behavior that is always too late or too early - “Shakey” the robot was shakey because he was constantly overreacting.

The problem with symbolically driven AI, is that every scenario must be anticipated and solved by hand: the designer has to effectively solve every conceivable real-world problem for the machine before the machine ever encounters it. The idea behind situated AI is to rather to build machines with robust *problem-solving behaviors*, the precise algorithmic characteristics of which remain unknown and irrelevant.

Biologically Plausible Intelligence

> Traditional AI has tried to demonstrate sophisticated reasoning in rather impoverished domains. The hope is that the ideas used will generalize to robust behavior in more complex domains. Nouvelle AI tries to demonstrate less sophisticated tasks operating robustly in noisy complex domains. The hope is that the ideas used will generalize to more sophisticated tasks.
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> *Elephants Don’t Play Chess*, §6.1

Symbolic AI is sometimes contrasted to the connectionist approach, as though a fundamental antagonism existed: but although the connectionist model attempts to mimic neuroplasticity, it’s subject to the same limitations of scope, because it still approaches intelligence as a *topdown* optimization problem.

Since the mid-2000s, connectionist AI and machine learning have garnered some impressive accomplishments and attention: image and speech recognition has approached useful levels, the game of Go has been mastered by a machine, etc. But the same criticisms apply: these are extremely limited domains, with well-defined parameters, and the intelligence achieved is by virtue of *optimized brute force* ultimately enabled by hardware improvements, not refinement of method. The search and backpropagation techniques pioneered in the 1960s and 1970s are still used, but driven by computational power at orders of magnitude greater scale: this kind of intelligence is so inefficient as to be biologically implausible, if not impossible. An organism does not learn with extravagant expenditures of repetition and backpropagating tweaks to a neural network: a child does not require 1 million repetitions of a word before recognizing it in the many different timbres of voice and accent, nor to see 100,000 images of a cat before internalizing the gestalt called “cat”, despite differences of color and context.

Principles

To distill the principles of situated AI:

1. Complex behavior may be a result of a complex environment, not a complex control system.
2. Relational maps are less prone to error than global.
3. The world is complex and textured, not polyhedral.
4. Robust real-world testing is required from the beginning.
5. Calibration must be continuous.

The resulting architecture looks like this:

1. Multiple asynchronous processors.
2. Concurrent operation of multiple overlapping goals.
3. Coordination between modules is achieved via selective monitoring and interference of data streams.

While it's indisputable that this design more closely resembles biological systems, the question remains: are our current computing technologies ready to emulate the *advantages* of this architecture, or are we merely wasting our time imitating the organic with the inorganic? Put another way, can the *nonlinear dynamics* of asynchronous modular computation be extracted from an organic context and sufficiently emulated with integrated circuitry? This is the question our century may eventually answer.